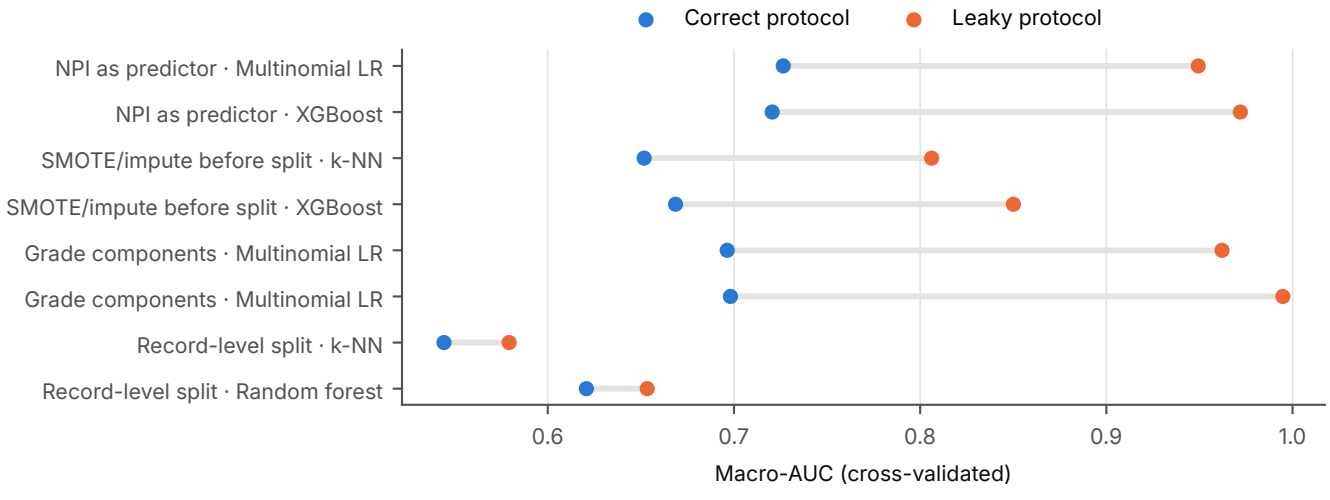
Correct protocol
Leaky protocol
NPI as predictor · Multinomial LR
NPI as predictor · XGBoost
SMOTE/impute before split · k-NN
SMOTE/impute before split · XGBoost
Grade components · Multinomial LR
Grade components · Multinomial LR
Record-level split · k-NN
Record-level split · Random forest
0.6
0.7
0.8
0.9
1.0
Macro-AUC (cross-validated)